Practical No : 4

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Roll No: 20

Problem statement :
Write a program to solve a 0-1 Knapsack problem using dynamic programming or branch and bound strategy.

Code :

```
def knapsack(capacity, weights, values, n):
    # Create a DP table
    dp = [[0 for _ in range(capacity + 1)]
          for _ in range(n + 1)]

    # Fill the DP table
    for i in range(1, n + 1):
        for w in range(1, capacity + 1):
            if weights[i - 1] <= w:
                dp[i][w] = max(
                    values[i - 1] + dp[i - 1][w - weights[i - 1]],
                    dp[i - 1][w]
                )
            else:
                dp[i][w] = dp[i - 1][w]

    return dp[n][capacity]


n = int(input("Enter number of items: "))

weights = []
values = []

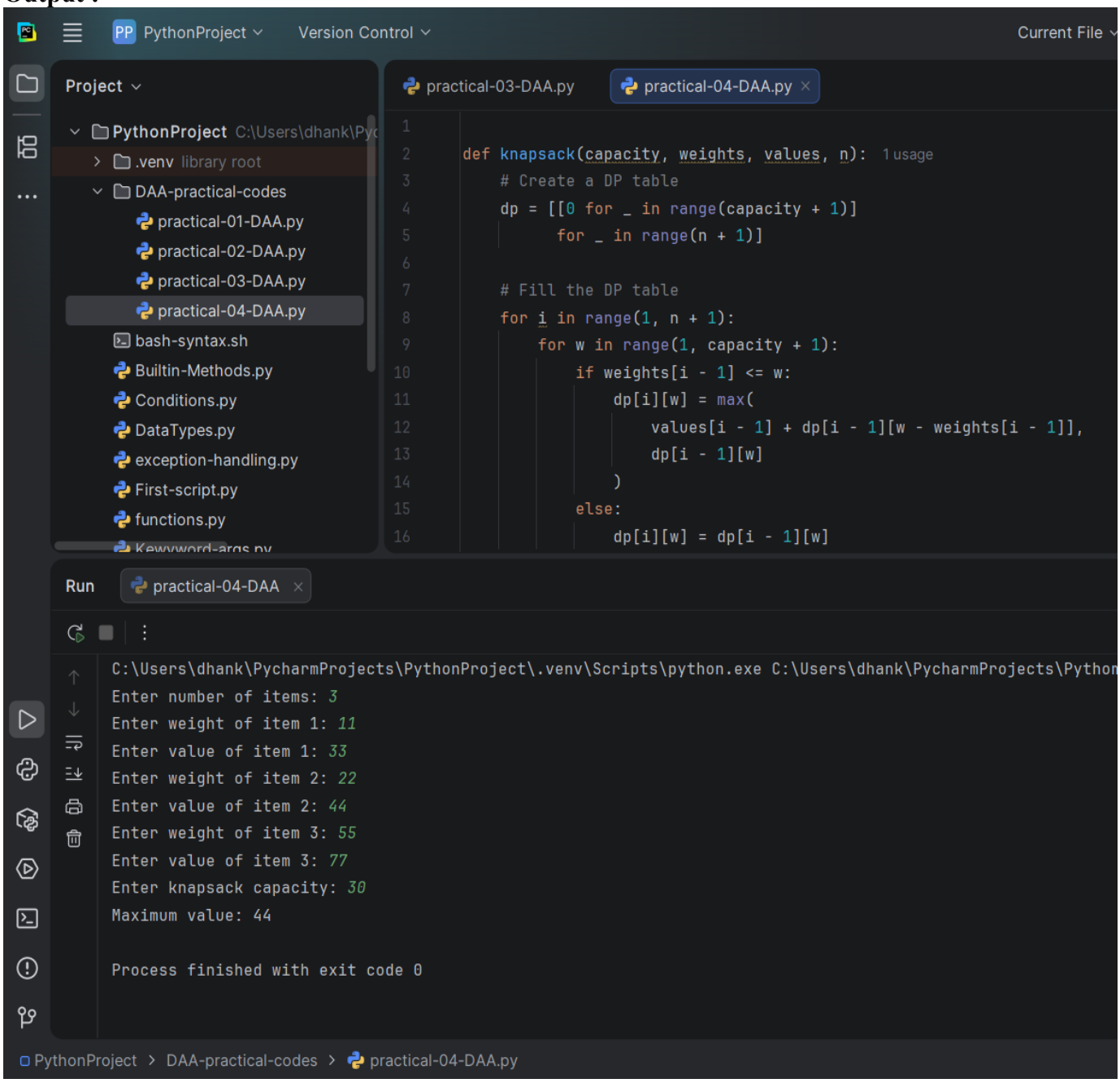
for i in range(n):
    weights.append(int(input(f"Enter weight of item {i + 1}: ")))
    values.append(int(input(f"Enter value of item {i + 1}: ")))

capacity = int(input("Enter knapsack capacity: "))

max_value = knapsack(capacity, weights, values, n)

print("Maximum value:", max_value)
```

Output :


PythonProject
Version Control
Current File
Project
PythonProject C:\Users\dhank\Pyc
.venv library root
DAA-practical-codes
practical-01-DAA.py
practical-02-DAA.py
practical-03-DAA.py
practical-04-DAA.py
bash-syntax.sh
Builtin-Methods.py
Conditions.py
DataTypes.py
exception-handling.py
First-script.py
functions.py
practical-03-DAA.py
practical-04-DAA.py
def knapsack(capacity, weights, values, n): 1 usage
Create a DP table
dp = [[0 for _ in range(capacity + 1)]
for _ in range(n + 1)]
Fill the DP table
for i in range(1, n + 1):
for w in range(1, capacity + 1):
if weights[i - 1] <= w:
dp[i][w] = max(
values[i - 1] + dp[i - 1][w - weights[i - 1]],
dp[i - 1][w]
)
else:
dp[i][w] = dp[i - 1][w]
Run
practical-04-DAA
C:\Users\dhank\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\dhank\PycharmProjects\Python
Enter number of items: 3
Enter weight of item 1: 11
Enter value of item 1: 33
Enter weight of item 2: 22
Enter value of item 2: 44
Enter weight of item 3: 55
Enter value of item 3: 77
Enter knapsack capacity: 30
Maximum value: 44
Process finished with exit code 0
PythonProject
DAA-practical-codes
practical-04-DAA.py